



DISRUPTIVE INNOVATIVE SOLUTIONS

TYLERSOFT BILLER ENGINE API DOCUMENT

Version 1.3

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Purpose

This document is intended for both technology personnel of Tylersoft-Electrics partner institutions. A basic knowledge of JSON format is assumed. Tylersoft Biller Engine API allows developers to connect securely to the Biller Engine Payment System to carryout utilities payments as well as airtime top up.

Revision History

Version	Review date	Author	Description
1.0	24/04/2018	Charles Mutura	Initial draft
1.1	04/10/2019	Gaonyadiwe Gaboiphiwe	Added sections for the callback and the message flow
1.2	20/08/2021	Gaonyadiwe Gaboiphiwe	Updated the sample callback message for BPC electricity. Added sections for the electricity customer confirmation and BPC transaction status query
1.3	13/09/2022	Gaonyadiwe Gaboiphiwe	Added the DSTV account confirmation section

List of SERVICES

Service ID	Biller name	Comment
1001	DSTV	DSTV
1002	MASCOM	Prepaid Airtime Top up
1003	BEMOBILE	Prepaid Airtime Top up
1004	ORANGE	Prepaid Airtime Top up
1005	BPC	BPC (Botswana power Corporation)
1006	INTERCAPE	Bus ticketing

Confirm BPC Customer

This applies to BPC transactions only

Endpoint: https://IP:port/TYLERSOFT_VPS_ADAPTOR/Confirm_Customer

Request format: JSON

Method: POST

Sample Request

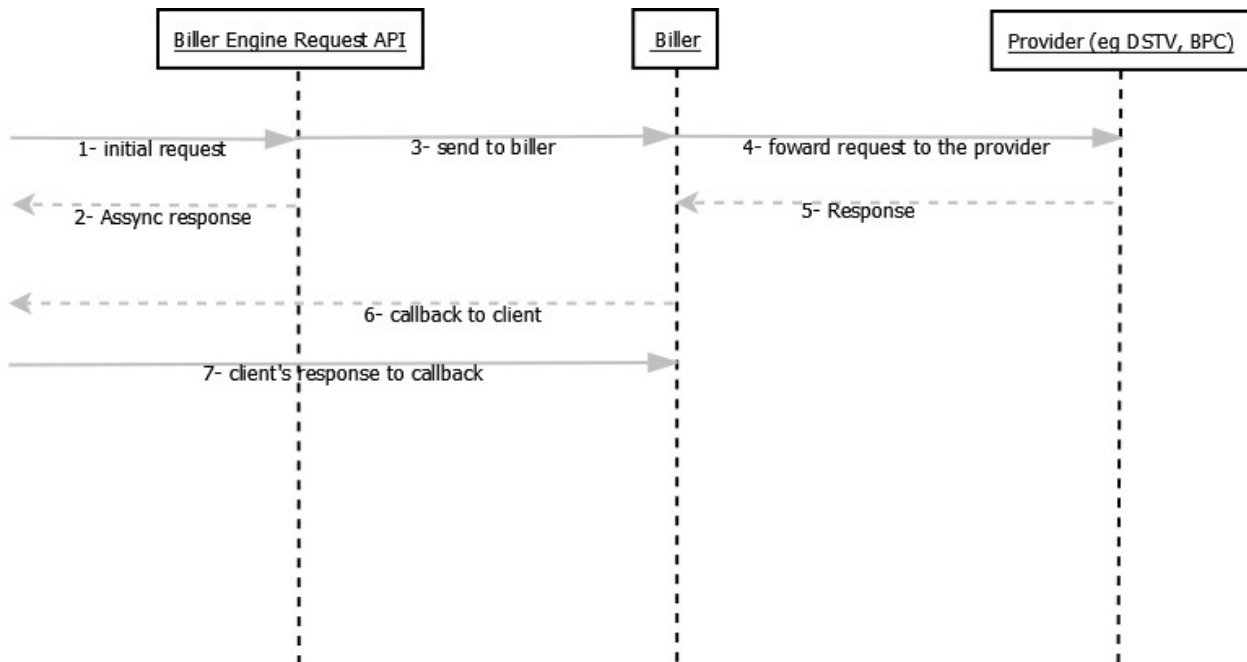
```
{
  "meternumber": "04040404040",
  "ClientId": "1005",
  "username": "testClient",
  "password": "clientpass"
}
```

Sample Response

```
{
  "maxvendamount": "0",
  "canvend": "true",
  "meternumber": "04040404040",
  "kyc": {
    "address": "",
    "contact": "",
    "name": "Test Vend",
    "utilitytype": "Electricity",
    "dayslastpurchase": "0"
  },
  "minvendamount": "0",
  "status": "00"
}
```

Send transaction for processing

Message flow diagram



Endpoint: https://IP:port/TylerSoftBillerEnine_Servlet/Servlet

Request format: JSON

Method: POST

Request Parameters

Parameter	Description	Optional/Mandatory
Amount	Total amount to pay for a service	M
Currency	Money currency used per country	M
ClientId	This is unique id per client -provided by Tylersoft	M
Serviceid	This identifies the biller e.g 1005 = BPC	M
Phonenumber	Customer phone number	O
accountno	e.g dstv decoder number, BPC meter number	M
username	Username of the source client	M
password	Password of the source client	M
TransactionId	Unique transaction reference number	M
email	End customer Email address	O
names	End customer names	M

Sample Request

```
{
  "Amount": "11.00",
  "Currency": "BWP",
  "ClientId": "1",
  "Serviceid": "1005",
  "Phonenumber": "26774769141",
  "accountno": "04040404040",
  "username": "xxxxxxx",
  "password": "xxxxxxxxx",
  "TransactionId": "011217203-BPC45",
  "email": "gaonyadiwe@tylersoft.net",
  "names": "G Gaboiphiwe"
}
```

Sample Response

```
{
  "Message": "Transaction received",
  "Responsecode": "00",
  "TransactionID": "230418155351"
}
```

Callback

Used to send final status of the transaction to client. Clients will be required to an endpoint/URL where the final status of the transaction will be sent to. TransactionId is used to map the received request and the callback message sent to the client.

Sample Callback message for other services (Airtimes)

```
{
  "Message": "success-Successful",
  "Responsecode": "00",
  "TransactionId": "220720191339-116"
}
```

Sample Callback message for BPC transactions (For a single Token)

```
{
  "domesticCharge": "",
  "Message": "Credit Purchase Successful ",
  "meterNumber": "04040404040",
  "Responsecode": "00",
  "tariffIndex": "1",
  "businessCharge": "0",
  "vat": "2.06",
  "receipt": "29941",
  "units": "14.8 kWh",
  "levy": "0.74",
  "TransactionId": "011217203-BPC45",
  "token": "0404-0404-0400-0000-0148",
  "keyChangeToken1": "",
  "keyChangeToken2": ""
}
```

Sample Callback message for BPC transactions (For multiple Tokens)

```
{
  "domesticCharge": "",
  "Message": "Credit Purchase Successful ",
  "meterNumber": "04040404040",
  "Responsecode": "00",
  "tariffIndex": "1",
  "businessCharge": "0",
  "vat": "2.06",
  "receipt": "29941",
  "units": "14.8 kWh",
  "levy": "0.74",
  "TransactionId": "011217203-BPC45",
  "token": "0404-0404-0400-0000-0148",
  "keyChangeToken1": "0404-0404-0455-0000-2023",
  "keyChangeToken2": "0404-0134-0400-0000-2024"
}
```

Sample callback response message

Upon receiving the callback the client should reply with the sample message below.

```
{"Status": "Success", "TransactionID": "trx00123", "message": "Successful"}
```

BPC transaction status query

This can be used as an alternative to the Callback. Clients can query BPC transactions status in case they did not receive the transaction status via the callback service. The query status request should be sent at least a minute after initiating the transaction.

Endpoint: `http://IP:port/ TYLERSOFT_VPS_ADAPTOR/QueryStatus`

Request format: JSON

Method: POST

Sample Request

```
{
  "TransactionId": "011217203BPC45",
  "ClientId": "1005",
  "username": "testClient",
  "password": "clientpass"
}
```

Sample Response for a single token

```
{
  "domesticCharge": "",
  "Message": "Credit Purchase Successful ",
  "meterNumber": "04040404040",
  "Responsecode": "00",
  "tariffIndex": "1",
  "businessCharge": "0",
  "vat": "2.06",
  "receipt": "29941",
  "units": "14.8 kWh",
  "levy": "0.74",
  "TransactionId": "011217203-BPC45",
  "token": "0404-0404-0400-0000-0148",
  "keyChangeToken1": "",
  "keyChangeToken2": ""
}
```

Sample Response for multiple tokens

```
{
  "domesticCharge": "",
  "Message": "Credit Purchase Successful ",
  "meterNumber": "04040404040",
  "Responsecode": "00",
  "tariffIndex": "1",
  "businessCharge": "0",
  "vat": "2.06",
  "receipt": "29941",
  "units": "14.8 kWh",
  "levy": "0.74",
  "TransactionId": "011217203-BPC45",
  "token": "0404-0404-0400-0000-0148",
  "keyChangeToken1": "0404-0404-0455-0000-2023",
  "keyChangeToken2": "0404-0134-0400-0000-2024"
}
```




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RESPONSE CODES

Code	Description
00	Successful/True
99	Failed /False
20	Pending Processing